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TOO MUCH OF A GOOD THING: WHO WE FOLLOW, WHAT WE DO, AND HOW MUCH TIME WE SPEND ON SOCIAL MEDIA AFFECTS WELL-BEING

MELISSA HUNT, KATHERINE ALL, BRENNAN BURNS, AND KYLER LI
University of Pennsylvania

Introduction: Given previous experimental research on the benefits of limiting time spent on social media and correlational research linking active use to better well-being, we designed an experimental study to investigate the effects of limiting time and increasing active use on social media on well-being. **Method:** After completing a survey on who they followed on social media and undergoing a week of baseline monitoring, 88 undergraduates at a private university were randomly assigned to either limit Facebook, Instagram, Twitter, and Snapchat use to 30 minutes in total per day and increase their active use, just limit their use to 30 minutes per day, or continue to use social media as usual for three weeks. **Results:** The initial survey revealed that following more friends was negatively correlated with loneliness whereas following more strangers was positively correlated with depression. Highly depressed participants in the limited use group showed significant reductions in depression compared to the control group, but the active group did not show similar benefits. Highly active participants in the active group reported greater loneliness and anxiety and lower self-esteem at week 4 compared to less active participants in the active group, whereas the most passive participants in the control group showed the worst depression overall. **Discussion:** Our findings suggest that following friends rather than strangers and limiting time spent on social media may lead to significant improvements in well-being, and that moderately active engagement may be the most adaptive.

Keywords: social media, Facebook, Snapchat, Instagram, Twitter, well-being, depression, loneliness, friends

Address correspondence to Melissa Hunt, 425 S University Ave., Levin Building, Philadelphia, PA 19104; E-mail: mhunt@psych.upenn.edu.

The use of social networking sites (SNS) is widespread throughout America, particularly among adults. As of February 2019, 72% of Americans use at least one form of social media; this number increases to 90% for U.S. adults between ages 18–29 (Social Media Fact Sheet, 2019). Facebook is currently the most popular SNS overall, but social media platforms such as Instagram, Twitter, and Snapchat continue to increase in popularity and are the most popular sites among teenagers and young adults (GenZ or the iGen; Smith & Anderson, 2018). As the use of SNS increases, so too do concerns about the potential negative effects of SNS on people's mental health and well-being.

A large number of studies have examined these effects, with a particular focus on Facebook. Researchers have found both direct and indirect positive correlations between self-reported Facebook use and depressive symptoms (Steers et al., 2014; Tandoc, Jr. et al., 2015), low self-esteem (Kalpidou et al., 2011), loneliness (Song et al., 2014), and low life satisfaction (Kross et al., 2013). Self-reported Instagram use is directly and indirectly positively correlated with depressive symptoms among young adults (Donnely & Kuss, 2016; Lup et al., 2015).

The specific ways people use SNS moderate their effects. The intensity of usage matters; adolescents who engage in light use of SNS, between .5 and 2 hours, appear to experience the highest levels of well-being compared to both non-users and heavy users (Twenge, 2019). Tromholt (2016) found that light Facebook users experienced no positive effects of quitting Facebook, as opposed to medium and heavy users who improved in well-being. Who people engage with on SNS is also important. The indirect association between Instagram use and depressive symptoms found by Lup et al. (2015) was significantly moderated by the number of strangers followed, with more strangers being positively associated with more depressive symptoms.

Social networking sites use can also be defined as active or passive. Active use involves producing content and communicating with others, such as posting a photo, sharing a status, or commenting on another person's post (Verduyn et al., 2015). Passive use involves consuming content without directly interacting with others, such as scrolling through one's feed or viewing others' posts (Hanley et al., 2019). Passive use is associated with declines in subjective well-being (Verduyn et al., 2017), whereas active use is associated with improvements in subjective well-being

(Verduyn et al., 2017) and less loneliness (Burke et al., 2010). In an experimental study in which participants were prompted to use SNS more actively by posting more status updates on Facebook than they usually did in a week, participants experienced reductions in loneliness (Deters & Mehl, 2013). When participants were prompted to use Facebook actively or passively for ten minutes in-lab, passive participants experienced a drop in well-being at the end of the day (Verduyn et al., 2015). The positive effects of quitting Facebook for a week were more pronounced in passive users (Tromholt, 2016).

The vast majority of studies on the associations between SNS use and well-being rely on self-report data for SNS usage. This raises some questions about the validity of findings, as Sewall and colleagues (2019) found that iPhone users misestimated their weekly social media use by 16.6 hours and that correlations between usage and well-being were more significant for estimated usage than actual usage. Additionally, the use of correlational rather than experimental studies means that a direct causal relationship between SNS use and well-being often cannot be established.

To address the issue of causality, studies have begun to examine the effects of experimentally limiting SNS use on well-being. Findings thus far have been mixed. In a study in which 1,300 Danish people were assigned to stop using Facebook for a week or continue using it as usual, quitting Facebook led to increases in life satisfaction and more positive emotions (Tromholt, 2016). Similarly, college students who were assigned to refrain from using Facebook for 48 hours in one study and students who chose their own usage-reduction goal in another study demonstrated increased life satisfaction both during the cessation period and directly after the cessation period (Hinsch & Sheldon, 2013). In contrast, participants in a study by Vanman and colleagues (2018) reported lower life satisfaction after giving up Facebook for five days, but participants did experience lower levels of physiological stress measured using cortisol levels. Notably, all three of these studies relied on self-reported Facebook usage and had no objective ways of measuring whether participants truly quit using the platform. Hunt et al. (2018) found that limiting the use of Facebook, Instagram, and Snapchat to 30 minutes a day, measured objectively through screenshots of iPhone battery usage, led to reductions in loneliness and depressive symptoms.

However, other studies have failed to find positive effects of quitting SNS. Stieger and Lewetz (2018) found that participants who quit all forms of SNS for seven days experienced increases in cravings and boredom, although this may be more indicative of the addictive qualities of SNS than any negative effects of quitting. Similar to previous studies, there was not an objective measure of SNS usage. Hall et al. (2019) found no significant effects of quitting SNS for one to four weeks, and Hanley et al. (2019) found that a one week vacation from Facebook and Instagram had no significant effect on passive users and resulted in a lower positive affect for active users. Hall et al. (2019) measured compliance by checking participants' activity on their social media platforms, although this method cannot detect passive usage, and Hanley et al. (2019) measured usage amount through RescueTime software, demonstrating an increased focus in the field on objective measures of SNS use.

With the exception of Hunt et al. (2018) and Hinsch and Sheldon (2013), all of these experimental studies involved quitting SNS completely rather than merely limiting usage. Keeping in mind the fact that light use appears to be more beneficial than heavy or no use and that SNS can be an important form of maintaining and building relationships (Burke et al., 2010), complete cessation would not appear to be the most helpful or realistic way of limiting SNS use. It is also worth noting that while studies such as Deters and Mehl (2013) and Verduyn et al. (2015) have looked at how active versus passive use affects well-being experimentally, neither looked at limiting use alongside changing use patterns on any platform besides Facebook, and only the Deters and Mehl (2013) study had an objective measure of activity.

Given these open questions, we designed a study based in large part on Hunt et al. (2018) to address potential weaknesses in past studies on these topics and provide further evidence on how social media may affect mental well-being. In order to look at how active and passive use affect well-being and how limiting social media use affects well-being, this experimental study includes 3 randomly assigned groups, including a control group, a limited use group, and a limited use group that was also instructed to use social media more actively. All groups participated in a 4-week study where social media use was measured objectively through the screen time app included in iPhones, including a 3-week intervention period for the limited

and active groups. Additionally, to get an idea of how who one follows affects well-being, the first and final surveys included questions about how many followers participants had and who they followed.

With previous research on these topics and the design of our study in mind, our hypotheses were as follows:

1. We expected that individuals in the limited condition who were highly depressed at baseline would report higher levels of well-being at week 4 than individuals in the control condition, as was found in Hunt et al. (2018).
2. Due to previous findings on the benefits of active social media use, we also expected that individuals in the active condition who were highly depressed at baseline would report higher levels of well-being at week 4 than individuals in the control or limited conditions who were highly depressed at baseline.
3. Finally, we expected to find a positive correlation between the number of friends followed per platform at baseline and baseline well-being, and a negative correlation between the number of strangers followed per platform at baseline and baseline well-being.

METHODS

PARTICIPANTS

A total of 151 undergraduates (110 women, 41 men) at a private mid-Atlantic university completed the initial baseline survey. However, due to participant withdrawal (36 participants), study noncompliance (13 participants), COVID-19 complications (12 participants), participants belonging to a different population (1 participant reported being an Instagram influencer), and onboarding error (1 participant), 88 participants (65 women, 23 men) completed all four weeks of the study. Forty-two participants completed the study in the fall semester, and 46 participants completed in the spring. Participants began the study on a rolling basis. The subject pool consisted of undergraduates enrolled in psychology courses for which they could participate

in studies to earn course credit. Participants were required to have an iPhone and an existing account on at least one platform out of Facebook, Snapchat, Instagram, and Twitter.

PROCEDURE

All participants were recruited using the Department of Psychology's undergraduate research participation system. Upon signing up, participants were directed to a secure Qualtrics platform where they consented to the study and completed the baseline survey of estimated usage on Facebook, Instagram, Snapchat, and Twitter, how many people they followed and were followed by on these platforms, and completed mood and well-being measures. Participants were also told to estimate how many of the people they followed on each platform were friends, acquaintances, or strangers.

Participants were then sent a welcome email describing the study in more detail and study procedures for the following four weeks. They were told that at the end of the baseline week, they would be emailed with their random assignment to one of three groups. Participants were instructed to use social media as they normally would for the next week and to fill out a daily survey reporting the number of actions they completed on Facebook, Snapchat, Twitter, and Instagram each day. Participants were told to use Facebook and Twitter only on their phones, not their computers, in order to ensure proper tracking of their social media use. (Instagram and Snapchat can only be accessed as smart phone apps.)

After completing the baseline week, participants were randomly assigned using Random.org to one of three groups and given instructions for their group. Group 1, referred to as the control group, would continue to use social media as they usually do. Group 2, the limited group, would be limiting activity on Facebook, Instagram, Snapchat, and Twitter to 30 minutes per day across all apps. Group 3, the active group, would be limiting activity on Facebook, Instagram, Snapchat, and Twitter to 30 minutes per day across all apps, as well as performing one action for every three minutes spent on social media. Actions were defined as posting, commenting, tagging users, posting

to their story, replying to a story/participating in a poll, tweeting something, or replying to a tweet. For Snapchat, participants were instructed to spend at least 75% of their time on the app producing content. Producing content included sending snaps, posting to their story, or replying to others' stories. Participants continued to complete a survey every day reporting the number of action items they completed on each social media platform. Every week, participants completed a survey with the same mood and well-being measures as the baseline survey. Participants also sent in screenshots of their screen time usage for their social media apps. At the end of the fourth week, participants completed the week 4 survey, which asked about their estimated social media usage, which social media platforms they used, how many people they followed or were followed by on each social media platform, and mood and well-being measures. Participants were told to estimate how many of the users they follow on each platform were friends, acquaintances, or strangers. The week 4 survey allowed participants to write in responses as to how the study impacted them and their social media use. After participants had completed the week 4 survey and sent in their weekly screenshot, they were sent a wrap-up email explaining that they would be receiving course credit shortly and that they were done with the study.

MEASURES

Fear of Missing Out. The Fear of Missing Out Scale (FoMOs; Przybylski et al., 2013) is a validated measure of distress related to missing out on social experiences ($\alpha = .87$). It consists of 10 items scored on a scale of 1 (not at all true of me) to 5 (extremely true of me); items include statements such as "I get anxious when I do not know what my friends are up to," "Sometimes I wonder if I spend too much time keeping up with what is going on," and "I fear others have more rewarding experiences than me."

Loneliness. The UCLA Loneliness Scale (revised UCLA Loneliness Scale; Russell et al., 1980) measures perceived social isolation. The original version was revised to include reverse-scored items and consists of 20 items, scored on a scale of 1 (never) to 4 (often). Sample items include statements such as "No one really knows me well," "My interests and ideas are not shared by those

around me," and "I feel in tune with the people around me" (reverse scored). The scale has good construct validity and internal consistency with $\alpha = 0.94$ (Russell et al., 1980).

Anxiety. The Spielberger State Anxiety Inventory (STAI-S; Spielberger et al., 1983) is a widely used measure of anxiety symptoms. The inventory consists of two instruction sets, which measure state (in-the-moment) and trait (general) anxiety. We only used the state anxiety version, which consists of 20 items such as "I feel worried" and "I feel calm" (reverse scored). Participants can respond on a scale of 1 (not at all) to 4 (extremely so). The state version of the scale has good internal consistency with $\alpha = 0.92$ (Spielberger et al., 1983).

Depression. The Beck Depression Inventory (BDI-II; Beck et al., 1996) is a standard clinical measure of depressive symptoms. It consists of 21 items covering the vegetative, affective, and cognitive symptoms of depression. Respondents can indicate the severity of each symptom on a scale of 0–3 (e.g., for the symptom loss of pleasure, one can respond: "I get as much pleasure as I ever did from the things I enjoy," "I do not enjoy things as much as I used to," "I get very little pleasure from the things I used to enjoy," or "I cannot get any pleasure from the things I used to enjoy"). The BDI-II has been shown to have good convergent and discriminant validity in a prior sample of college students, as well as good internal consistency with $\alpha = .89$ (Steer & Clark, 1997).

Self-Esteem. The Rosenberg Self-Esteem Scale (RSES; Rosenberg, 1979) assesses how one feels about oneself. It consists of 10 items, scored 0 (strongly disagree) to 3 (strongly agree), with higher scores indicating more positive feelings about oneself. Items include "I feel that I have a number of good qualities," "I feel I do not have much to be proud of" (reverse scored), and "I take a positive attitude toward myself." The RSES has good internal consistency across multiple demographic groups with $\alpha = .91$ (Sinclair et al., 2010)

Social Comparison. The Iowa-Netherlands Comparison Orientation Measure (INCOM; Gibbons & Buunk, 1999) measures one's tendency to engage in social comparisons. It consists of 11 items, scored 1 (disagree strongly) to 5 (agree strongly), with higher

scores indicating stronger tendencies to engage in social comparison. The items cover social comparisons regarding ability (e.g., I always pay a lot of attention to how I do things compared with how others do things) and opinion (e.g., I often like to talk with others about mutual opinions and experiences). The scale has good internal consistency with α ranging from .78 to .85 across 10 different American samples (Gibbons & Buunk, 1999).

SOCIAL MEDIA USAGE

Participants were asked to estimate their daily usage of Facebook, Instagram, Snapchat, and Twitter before starting the baseline week. To get objective data on social media during the baseline week and following weeks, we had participants email screenshots of their iPhone Screen Time at the end of every week. iPhones automatically track the total minutes each application is actively open on the screen. The Screen Time usage allows users to display their usage for the past 24 hours or 7 days. We provided instructions on how to get the Screen Time usage screen in our initial welcome email.

To track passive/active social media use, participants completed a survey every night reporting the number of actions they completed on Facebook, Snapchat, Twitter, and Instagram. Reminder e-mails were sent every night at 9PM along with a link to the survey.

RESULTS

BASELINE RESULTS

Correlational Results for People Followed and Well-Being. The number of people that participants followed and considered a friend was negatively correlated with loneliness ($r = -.26$, $p < .01$) and positively correlated with social comparison ($r = .26$, $p < .001$). When social comparison was controlled for, the negative correlation between loneliness and friends followed increased to $r = -.30$ ($p < .01$). The number of strangers that participants followed, on the other hand, was positively correlated with depression ($r = .20$, $p < .05$), Fear of Missing Out ($r = .20$, $p < .05$), and social comparison ($r = .23$, $p < .01$; see Table 1). Thus, it appears

TABLE 1. Correlations for Types of People Following and Well-Being.

	#Friends Following	#Strangers Following	Loneliness	Depression	Fear of Missing Out	Social Comparison
1. #Friends Following						
2. #Strangers Following	.175*					
3. Loneliness	–.264**	.117				
4. Depression	–.085	.204*	.594**			
5. Fear of Missing Out	.113	.195*	.289**	.425**		
6. Social Comparison	.261**	.228**	.097	.331**	.569**	

*$p < .05$; **$p < .01$.

that following more people overall is associated with greater social comparison; however, following more friends is associated with less loneliness and following more strangers is associated with greater depression and Fear of Missing Out.

EXPERIMENTAL RESULTS

Manipulation Check. First, we ensured that participants in the limited and active conditions decreased their time spent on social media. During the baseline week before assignments were announced, the time spent on social media was similar for participants in the control (M = 471, SD = 333), limited (M = 412, SD = 271), and active (M = 466, SD = 261) groups. Interestingly, people were not very good at estimating how much time they were spending on social media. The overall correlation between estimated use and actual use was significant, but modest (r = .23, $p < .05$). Surprisingly, people tended to overestimate how much time they spent. On average, before starting the baseline week, people estimated that they used about 122 minutes (SD = 78), or two hours of social media a day. In fact, they used about an hour a day (M = 446, SD = 284) while engaging in active self-monitoring. Over the next three weeks, the participants in the limited group, t(4.04), $p < .001$, and the active group, t(5.73), $p < .001$, spent significantly less time on social media than the control group. As expected, time spent on social media did not differ significantly between the limited and active groups (see Figure 1).

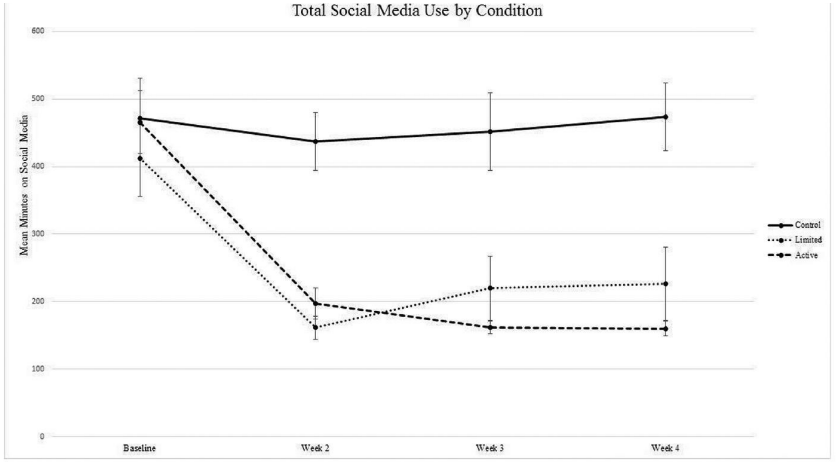


FIGURE 1. Total Social Media Use by Condition.

In order to ensure that participants in the active group increased their active use of social media, we calculated *z*-scores for the activity level of participants based on the number of actions per minute and the percentage of time that participants reported being active on Snapchat. At baseline, the three groups did not differ significantly in activity level. Over the next three weeks, the active group reported significantly higher activity levels than the control group, $t(9.73)$, $p < .001$, and the limited group, $t(6.72)$, $p < .001$. The limited and control groups did not differ significantly in terms of activity level (see Figure 2).

Effect of Condition on Depression. Replicating the core finding of Hunt et al. (2018), there was a significant interaction effect between baseline depression and condition, $F(2, 87) = 3.081$, $p = .05$, such that those individuals who reported more depressive symptoms at baseline showed significantly less depression at follow-up when they limited their SNS use than when they did not. Highly depressed participants in the limited group reported decreases in depression (at baseline, mean BDI = 19.80, $SD = 2.89$ at week 4, mean BDI = 14.80, $SD = 5.41$) and reported significantly less depression than those in the control group (at

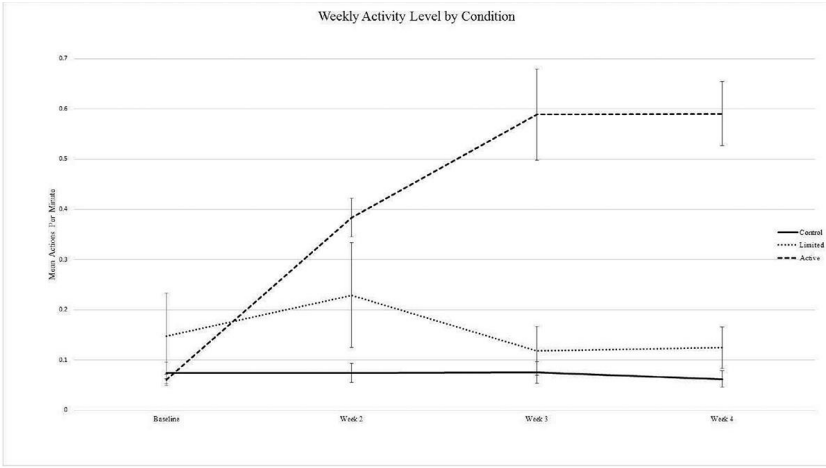


FIGURE 2. Weekly Activity Level by Condition.

baseline, mean baseline BDI = 24.00, *SD* = 2.21, at week 4, mean BDI = 28.63, *SD* = 5.04) (see Figure 3).

However, contrary to our expectations, highly depressed participants in the active group showed no such improvements. While they still reported lower levels of depression at week 4 compared to the control group, they did not significantly decrease in depression over time (at baseline, mean BDI = 17.29, *SD* = 2.83, at week 4, mean BDI = 16.29, *SD* = 5.22; see Figure 4). In other words, highly depressed participants who limited their time on social media experienced reductions in depressive symptoms from baseline to week 4. Participants who limited their time on social media and increased their activity levels did not experience reductions in depressive symptoms, but they still reported less depressive symptoms at week 4 than the control group.

Effect of High Activity Levels on Well-Being. Contrary to our hypotheses, participants in the active condition experienced less improvement in well-being than those in the limited condition. To better understand these results, we took a closer look at the activity levels of participants in the active condition. We divided participants in the active condition based on a median split of activity level z-scores, thus allowing us to compare individual

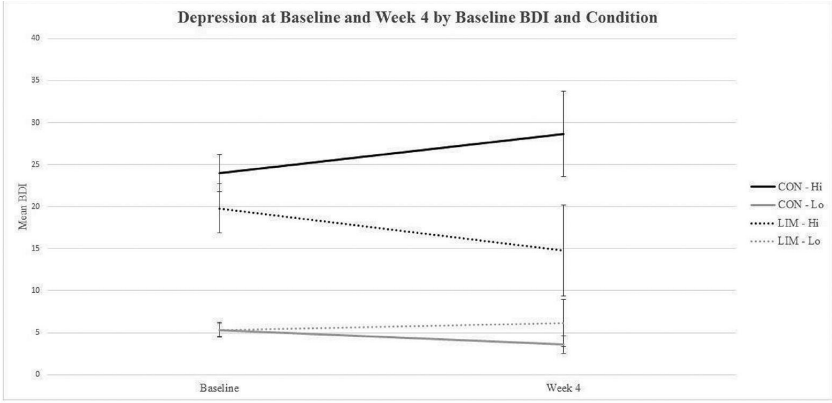


FIGURE 3. Depression Scores at Baseline and Week 4 by Condition and Baseline Depression.

differences in the degree to which participants increased their activity level. Highly active participants in the active condition reported greater loneliness, $t(-2.74)$, $p = .01$, greater anxiety $t(-2.29)$, $p < .05$, and lower self-esteem $t(2.44)$, $p < .05$, than less active participants. These findings were trending for greater depression $t(-2.02)$, $p = .052$, and social comparison $t(-1.866)$, $p = .07$ (see Figure 5). In other words, participants in the active group who increased their activity to a greater degree reported worse well-being at week 4. We repeated these procedures with the limited and control groups by dividing the participants into high and low activity based on positive or negative activity z-scores. We found no significant differences between the high activity and low activity groups, implying that participants in the active group increased their activity level to an unnatural degree which led to the adverse effects of high activity.

In order to further investigate the effects of high activity in the active group, we ran an ANOVA to assess the effect of activity level on depression at week 4, controlling for high or low baseline depression. This analysis revealed a significant main effect of both activity level, $F(1, 32) = 12.11$, $p < .01$, and baseline depression, $F(1, 32) = 9.02$, $p < .01$, on depression at week 4, as well as a significant interaction effect between the two, $F(1, 32) = 8.40$, $p < .01$. Participants who were highly depressed at baseline and

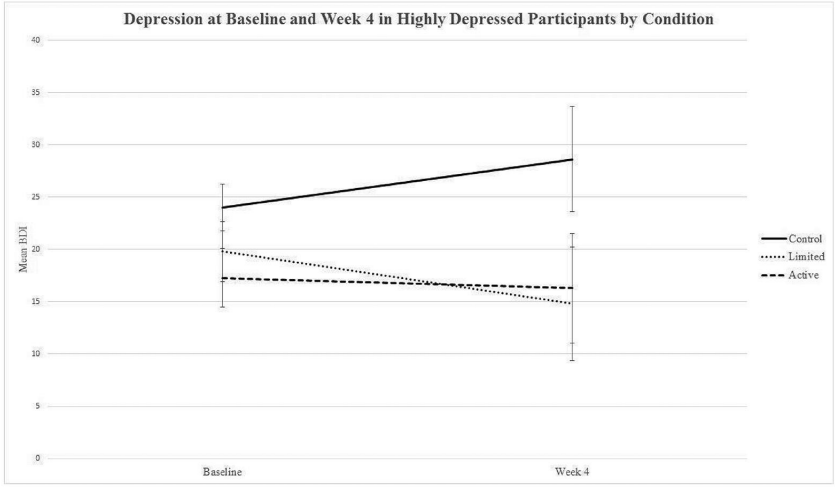


FIGURE 4. Depression at Baseline and Week 4 in Highly Depressed Participants by Condition.

highly active reported far greater depression at week 4 (Mean BDI = 25.67, *SD* = 3.38) than those who were less active (Mean BDI = 3.00, *SD* = 5.89). We then ran a similar ANOVA for the control group using the same variables. This analysis also revealed a significant main effect of both activity level, $F(1, 35) = 7.58$, $p = .01$, and baseline depression, $F(1, 35) = 69.45$, $p < .001$, on depression at week 4, as well as a significant interaction effect between the two, $F(1, 35) = 13.38$, $p = .001$. However, the effect of activity level was in the opposite direction. Highly depressed participants who were also more active reported lower levels of depression (Mean BDI = 18.40, *SD* = 3.24) than those who were less active (Mean BDI = 36.50, *SD* = 3.62, see Figure 6). A similar ANOVA for the limited condition revealed no significant results.

In other words, highly depressed participants who were naturally more active in the control condition reported lower levels of depression at week 4, but highly depressed participants who were more active in the active condition reported greater levels of depression. It is important to note that highly active participants in the control condition were still far less active than highly active participants in the active condition, as participants in the

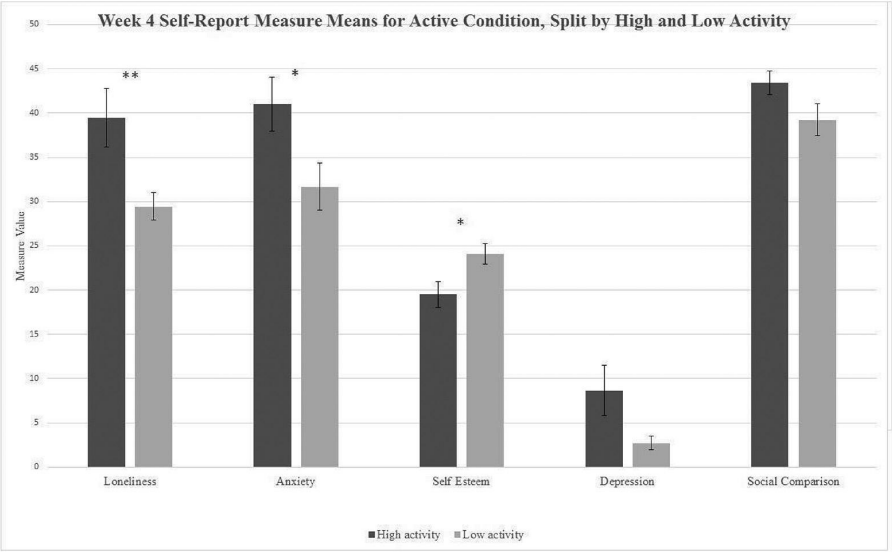


FIGURE 5. Effect of High Activity Levels on Well-Being.

active group were divided using a median split since almost all of their activity *z*-scores were much higher than zero. Thus, it appears that high activity only begins to have negative effects when participants are already limiting their time on social media and increase their activity to an unnatural degree. When participants use social media without limit, then being more active is associated with better well-being. Notably, baseline depression did not predict activity levels in the following week, so it is not the case that more depressed participants are simply more or less active.

DISCUSSION

Our hypotheses regarding correlations between well-being and who participants followed on social media were confirmed. Baseline data showed a significant negative correlation between the number of friends followed on social media and loneliness and a significant positive correlation between the number of

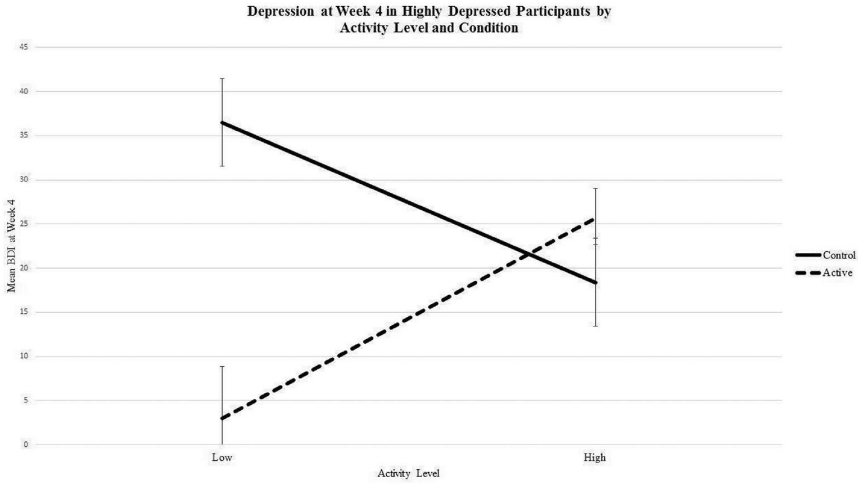


FIGURE 6. Depression at Week 4 in Highly Depressed Participants, Split by Activity Level and Condition.

strangers followed and baseline depression and FOMO. Both number of friends followed and number of strangers followed were significantly correlated with increased social comparison, which moderated the correlation between friends followed and loneliness. Therefore, who people follow on social media appears to be an important factor when assessing social media's impact on well-being. Following more friends could allow social media users to feel more connected with others (Ellison et al., 2007) and maintain social bonds across distances or in times of social isolation. Following more strangers, on the other hand, could present users with only the polished, seemingly perfect lives of others without the benefits of social connection.

Our hypotheses regarding our replication of Hunt et al. (2018) were partially confirmed. We replicated their central finding concerning participants who were highly depressed at baseline, in which participants in the limited group reported lower depression at week 4 compared to participants in the control group. However, these differences did not appear in non-depressed participants and we did not replicate a similar finding for loneliness. As such, limiting social media appears to be the most

beneficial for individuals who are already experiencing high levels of depression.

We also hypothesized that highly depressed participants in the active condition would report higher levels of well-being at week 4 than participants in the control and limited conditions. However, while these participants still reported lower levels of depression at week 4 compared to the control group, they did not experience any decreases in depression and appeared to fare worse than the limited group. In order to better understand this effect, we used a median split to divide participants in the active condition based on individual differences in the degree to which participants increased their activity levels. Participants in the active group who increased their activity to a greater extent reported worse loneliness, anxiety, and self-esteem than those who were less active. When we split the control group by activity level, on the other hand, highly depressed participants in the control group who were naturally more active reported decreased depression at week 4 compared to those who were less active.

The fact that participants in the control group who were naturally less active reported worse well-being supports previous findings on the negative effects of passive use. This could also explain why limiting social media is beneficial for highly depressed participants; while participants in the limited group did not increase their actions per minute on social media, they most likely reduced their amount of passive scrolling when they were limited to 30 minutes per day. This is consistent with previous findings that passive participants show more pronounced effects of limiting time spent on social media (Tromholt, 2016). We believe our study confirms the benefits of limiting passive use; however, the effects of increasing active use are less clear.

We considered two potential explanations for the negative outcomes of highly active participants within the active group. One possibility is that our guidelines for the study called for an unnaturally high level of activity when defining active use. As there were no previous quantitative definitions of active use, we constructed our definition of one action for every three minutes of use based on informal surveys of users' activity estimates. However, analyses of activity levels during the baseline week for all participants revealed that users perform closer to one action for every twelve minutes spent on social media. Thus, our study

may have required participants in the active group to increase their activity levels to an unnatural degree, therefore masking the true benefits of active use. Perhaps unsurprisingly, it may well be that moderately active use is the most adaptive approach.

Another possibility is that requiring participants to be active on social media when they would not otherwise choose to be is genuinely harmful to well-being. Participants may feel pressured to come up with content to post or feel insecure about interacting with friends on social media that they usually do not speak to. Additionally, increased communication brings with it the risk of increased conflict, which is further exacerbated by difficulties in communicating tone and intention over the Internet. If these posts on social media are not well-received, this could lead to regret and further decreases in well-being that would not have occurred had the user never posted. If this is the case, then future researchers must carefully consider the ethical implications of inducing increased activity and, possibly, increased regret and conflict on the part of participants. Our findings indicate that interventions related to social media use would perhaps best be served by focusing on reducing passive use and overall time spent rather than increasing active use.

LIMITATIONS

This study had several important limitations. While we asked participants for the number of following/followers they had on each social media platform, we lacked an objective way to verify the information they submitted. We also lacked an objective measure for friends/acquaintances/strangers estimates. Participants were instructed to identify the number of friends, acquaintances, or strangers out of the users that they followed on each social media platform. However, this categorization was largely subjective as exact definitions of what constitutes a friend or acquaintance may differ from participant to participant.

Furthermore, we lacked an objective measure for active use. While we had participants report their actions at the end of every day, we were not able to check whether or not they had completed the action they reported. The fact that these surveys were submitted daily makes forgetfulness less likely, but it is possible that participants in the active condition misreported

their number of daily activities in order to comply with study guidelines. However, considering we found significant differences in well-being that were associated with reported activity levels within the active condition, misreporting was unlikely.

Future studies would also benefit from using an objective measure of followers/following data and an objective measure of active activity. Participants may struggle to accurately estimate these numbers or otherwise misreport this information. This could be achieved through having users send in screenshots of their followers/following counts or following participants on social media using a dummy account to track actions. While this would still not capture all data, such as comments on others' photos or Snapchat information, it would be able to provide more objective information. Another option would be to experimentally limit the number of strangers participants follow to go beyond simple correlational data.

In addition, there was no empirically established benchmark for what qualifies as active use so we set the benchmark at one action for every three minutes spent on social media based on a small informal survey. However, based on our results, our benchmark may have been over the target zone. During the baseline week, participants performed closer to one action for every twelve minutes spent on social media. Thus, our requirements for the active condition most likely did not resemble genuine active use for most social media users, and instead pushed activity to an unnatural degree. Future studies could conduct more refined ecological analyses of how much active use is harmful versus beneficial. In our study, the active condition may have been *too* active. Future studies could focus on finding the ideal point where participants are engaging in enough active use to see benefits, but not so much active use that they are experiencing adverse effects. This could involve assigning varying levels of activity while participants are limiting social media usage or asking participants to use social media more actively without time restrictions. A better understanding of the effects of specific characteristics of social media could allow users to maximize the benefits of social connection while avoiding the declines in well-being usually associated with social media usage.

While we instructed participants to only use social media on their mobile phones, participants still had the ability to use social media on their or their friends' computers and other devices.

There is a possibility that participants' screen time screenshots may not reflect their true social media usage. In addition, because we were using iPhone screen time screenshots as an objective measure of social media use, the study sample was limited to students with iPhones.

Because we were relying on the app usage metrics available on iPhones, our findings may not generalize to Android users. iPhones are significantly more expensive than Android devices, and our sample may have been unusually wealthy, even for a population of college students at a private university. Future studies of this kind should include Android users. Additionally, studies that also track use on other devices, such as computers, could gain a more thorough understanding of how much social media participants are truly using.

Lastly, we had a smaller sample size than anticipated due to COVID-19. Many participants needed to use social media to stay in contact with their family and friends, so not all our participants were able to complete the study.

CONCLUSION

Prior research on the effects of social media is largely correlational in nature, focuses on only one platform, relies exclusively on self-report, or requires participants to completely eliminate social media usage, thus limiting its external validity. The extensive and detailed data we collected on the characteristics of participants' social media usage across multiple platforms allowed us to establish direct correlations between well-being and following friends and strangers on social media platforms. The use of multiple platforms and asking participants to limit rather than eliminate their use increased our study's ecological validity, thus allowing us to better understand how people can modify their social media usage in the real world in order to improve well-being. Our study was also the first that we know of to numerically quantify active use in a manner that allows for refined comparisons between platforms and participants. Through our experimental manipulation of activity, we have established a more nuanced understanding of when active use is genuinely beneficial and when it might push the boundaries into maladaptive. As social media platforms continue to grow and people around the world come to rely on

them to a greater extent for connection in general and in times of isolation brought about by COVID-19, studies such as this are imperative in order to inform our recommendations about the most adaptive way to use social media.

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